

NETAPP ONTAP VSCAN

Enterprise Lab Guide & Administration Handbook

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1. Introduction

What is Vscan?

Vscan (Virus Scan) is ONTAP's antivirus integration framework that allows external antivirus servers to scan files stored on NetApp SMB/CIFS shares.

ONTAP itself does not contain an antivirus engine.

Instead, ONTAP communicates with external antivirus software running on Windows Vscan Servers.

Supported vendors include:

- Trend Micro
 - Symantec
 - McAfee
 - Sophos
 - Other certified NetApp Vscan partners
-

Why Vscan?

Without Vscan:

```
Client --> SMB Share --> File Access Granted
```

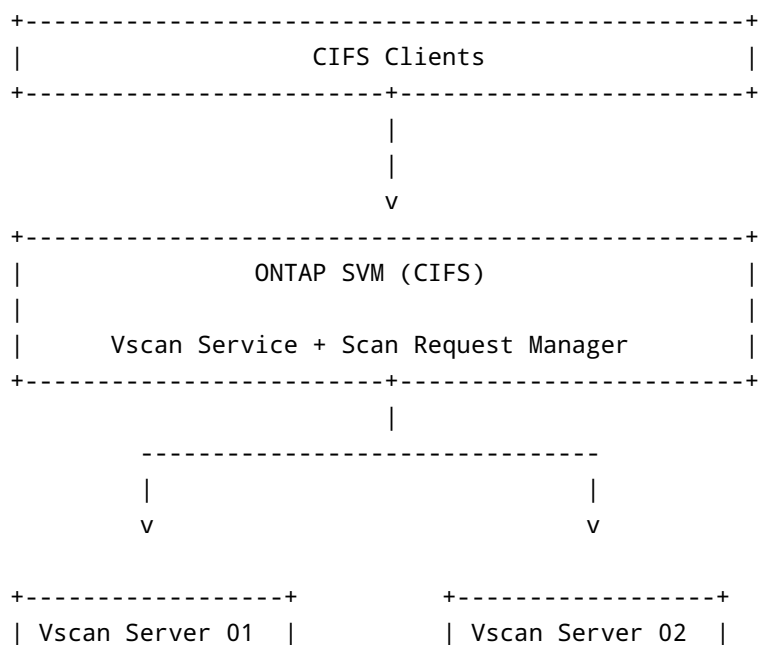
Potential malware may enter storage undetected.

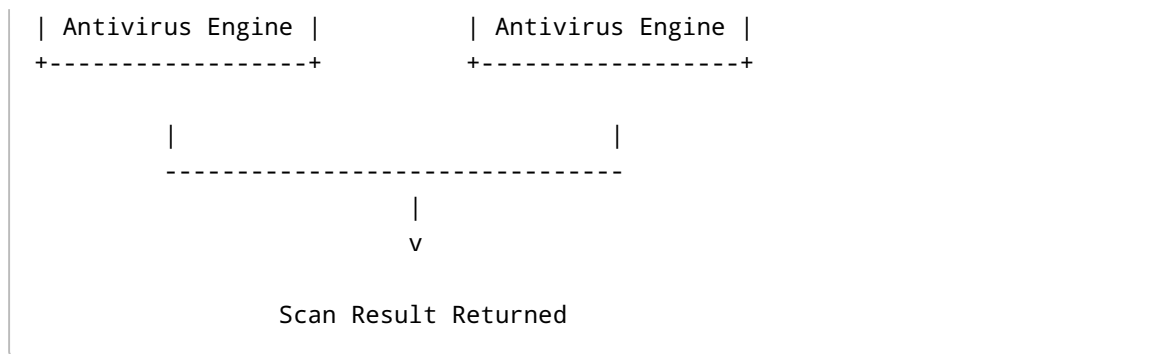
With Vscan:

```
Client --> SMB Share
      |
      v
    ONTAP Vscan
      |
      v
  Antivirus Server
      |
    Clean / Infected
      |
      v
    Allow / Block Access
```

2. Vscan Architecture

Enterprise Architecture





3. Components Overview

Vscan Service

Built-in ONTAP service responsible for:

- Scan request generation
- Scanner selection
- Communication with AV servers
- Scan result processing

Scanner Pool

Collection of approved antivirus servers.

Example:

POOL: VSCAN_POOL

Members:

10.10.10.11

10.10.10.12

10.10.10.13

Benefits:

- Load balancing
- Redundancy
- High availability

Privileged User

Special AD account used by antivirus servers.

Example:

```
DOMAIN\vscan_service
```

Requirements:

- Backup Operators
- Local Administrator
- Scan access permissions

On-Access Policy

Controls:

- Which files are scanned
- When files are scanned
- Which extensions are scanned
- Which paths are excluded

On-Demand Task

Used for:

- Scheduled scanning
- Full file system scans
- Compliance scans

4. How Vscan Works

File Open Flow

```
User Opens File
      |
      v
ONTAP Receives Request
      |
      v
Policy Match?
      |
```

```
    YES
    |
    v
Send Scan Request
    |
    v
Vscan Server Scans
    |
    v
Result Returned
    |
    Clean/Infected
    |
    v
Allow / Block
```

File Write Flow

```
File Created
    |
    v
Policy Triggered
    |
    v
Scan Request Sent
    |
    v
AV Server Validation
    |
    v
Result Returned
```

5. Scanner Pool Configuration

Step 1 Create Scanner Pool

```
vserver vscan scanner-pool create
-vserver svm_prod
-scanner-pool VSCAN_POOL
-privileged-users DOMAIN\vscan_service
```

Step 2 Add Servers

```
vserver vscan scanner-pool servers add
-vserver svm_prod
-scanner-pool VSCAN_POOL
-hostnames av01,av02,av03
```

Step 3 Verify

```
vserver vscan scanner-pool show
```

Expected:

```
Scanner Pool : VSCAN_POOL
Servers      : av01,av02,av03
```

Step 4 Apply Policy

```
vserver vscan scanner-pool apply-policy
-vserver svm_prod
-scanner-pool VSCAN_POOL
-scanner-policy primary
```

6. On-Access Policies

Create Policy

```
vserver vscan on-access-policy create
-vserver svm_prod
-policy-name PROD_POLICY
```

Enable Policy

```
vserver vscan on-access-policy enable
-vserver svm_prod
-policy-name PROD_POLICY
```

Show Policy

```
vserver vscan on-access-policy show
```

Add Included Extensions

```
vserver vscan on-access-policy file-ext-to-include add
-vserver svm_prod
-policy-name PROD_POLICY
-file-extensions exe,dll,zip,pdf
```

Add Excluded Extensions

```
vserver vscan on-access-policy file-ext-to-exclude add
-vserver svm_prod
-policy-name PROD_POLICY
-file-extensions log,tmp
```

Exclude Path

```
vserver vscan on-access-policy paths-to-exclude add
-vserver svm_prod
-policy-name PROD_POLICY
-paths /vol/backup
```

7. On-Demand Scanning

Create Task

```
vserver vscan on-demand-task create  
-vserver svm_prod  
-task-name WEEKLY_SCAN  
-paths /vol/data
```

Run Scan

```
vserver vscan on-demand-task run  
-vserver svm_prod  
-task-name WEEKLY_SCAN
```

Schedule Scan

```
vserver vscan on-demand-task schedule
```

Reports

```
vserver vscan on-demand-task report show
```

8. Complete Deployment Lab

Lab Topology

```
Cluster Name : cluster01
```

```
Node1 : node01
```

```
Node2 : node02
```

```
SVM : svm_prod
```

```
CIFS LIF :  
10.10.100.101
```


Vscan Servers:

10.10.50.11

10.10.50.12

Configuration Sequence

```
1 Create CIFS Server
  |
2 Create Scanner Pool
  |
3 Add AV Servers
  |
4 Configure Privileged User
  |
5 Create On-Access Policy
  |
6 Enable Policy
  |
7 Enable Vscan
  |
8 Validate Connectivity
  |
9 Test Scan
```

Enable Vscan

```
vserver vscan enable
-vserver svm_prod
```

Verify

```
vserver vscan show
```

Output:

```
Vserver      Vscan Status
-----
svm_prod     on
```

9. Monitoring & Validation

Show Status

```
vserver vscan show
```

Connection Status

```
vserver vscan connection-status show-all
```

Connected Servers

```
vserver vscan connection-status show-connected
```

Disconnected Servers

```
vserver vscan connection-status show-not-connected
```

Active Pools

```
vserver vscan scanner-pool show-active
```

Event Logs

```
vserver vscan show-events
```

Detailed Events

```
vserver vscan show-events -instance
```

10. Troubleshooting Scenarios

Scenario 1

Users Can Access Files But Scans Are Not Happening

Check:

```
vserver vscan show
```

Expected:

```
Status: on
```

If OFF:

```
vserver vscan enable -vserver svm_prod
```

Scenario 2

Antivirus Server Not Connecting

Check:

```
vserver vscan connection-status show-all
```

Possible Causes:

- DNS issue
- Firewall issue
- Privileged user issue
- Antivirus service stopped

Scenario 3

Scanner Pool Offline

Check:

```
vserver vscan scanner-pool show-active
```

If empty:

```
vserver vscan scanner-pool show
```

Verify:

- Servers added
 - Policy applied
 - Hostname resolution
-

Scenario 4

Infected File Not Blocked

Check:

```
vserver vscan show-events
```

Verify:

- Policy enabled
 - Extension included
 - AV signatures updated
-

Scenario 5

High CIFS Latency

Check:

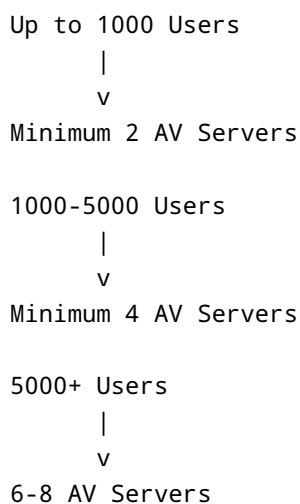
```
vserver vscan connection-status show-all
```

Possible Causes:

- Overloaded AV server
 - Too many file extensions
 - Large file scans
 - Network latency
-

11. Performance Considerations

Recommended Design



Best Practices

- ✓ Exclude log files
- ✓ Exclude backup folders
- ✓ Use multiple AV servers
- ✓ Use dedicated VLAN
- ✓ Schedule full scans after hours
- ✓ Monitor connection status

12. Security Best Practices

Recommended

- Use dedicated AD service account
 - Restrict scanner server access
 - Update AV signatures daily
 - Enable event monitoring
 - Maintain redundant scanner pools
-

13. Interview Questions

What is Vscan?

Vscan is ONTAP's antivirus integration framework that offloads virus scanning to external antivirus servers.

Does ONTAP Perform Antivirus Scanning?

No.

ONTAP sends scan requests to external Vscan servers.

Difference Between On-Access and On-Demand?

On-Access	On-Demand
Real-time	Scheduled
User-triggered	Admin-triggered
File open/write	Full scan

What Is Scanner Pool?

Logical collection of antivirus servers used by ONTAP.

Command To Verify Connectivity?

```
vserver vscan connection-status show-all
```

Command To View Infected Files?

```
vserver vscan show-events
```

Command To Reset Cached Scan Information?

```
vserver vscan reset -vserver svm_prod
```

14. Quick Command Cheat Sheet

Vscan Service

```
vserver vscan enable  
vserver vscan disable  
vserver vscan reset  
vserver vscan show
```

Events

```
vserver vscan show-events
```

Connection Status

```
vserver vscan connection-status show-all  
vserver vscan connection-status show-connected  
vserver vscan connection-status show-not-connected
```

Scanner Pool

```
vserver vscan scanner-pool create  
vserver vscan scanner-pool show  
vserver vscan scanner-pool modify  
vserver vscan scanner-pool delete
```

On-Access Policy

```
vserver vscan on-access-policy create  
vserver vscan on-access-policy show  
vserver vscan on-access-policy modify
```

On-Demand Tasks

```
vserver vscan on-demand-task create  
vserver vscan on-demand-task run  
vserver vscan on-demand-task show  
vserver vscan on-demand-task report show
```

Final Summary

A production-ready Vscan deployment consists of:

- ✓ Scanner Pool
- ✓ Privileged User
- ✓ Antivirus Servers
- ✓ On-Access Policies
- ✓ On-Demand Tasks
- ✓ Monitoring
- ✓ Event Analysis
- ✓ Performance Tuning

A healthy Vscan environment ensures secure CIFS access while maintaining enterprise-grade malware protection and compliance.